

2023 Team Description Paper: SPbUnited^{*}

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Abstract. We are team SPbUnited from Saint Petersburg University and Presidential lyceum of physics and mathematics №239 (Saint Petersburg, Russia). Our team is formed by former URoboRus members, so we changed name to represent this changes. We are going to refer to URoboRus tdp. We are participating in RoboCup Small Size League in 2023. This TDP provides a technical overview of our robots, control software system and algorithms.

Keywords: RoboCup · robotics · multi-agent system · hybrid centralized system

1 Introduction. Past experience

The core of our team consists of students of the Department of Theoretical Cybernetics of St. Petersburg State University and students of the Presidential Physics and Mathematics Lyceum No. 239. We work together with engineers who have developed robots and maintain them in working order. We started creating our solution in September 2018. By the end of January 2019, the team had qualified for Robocup 2019. In 2020, we also qualified, and in June 2021, we took part in the RoboCup SSL virtual competitions. Also in October 2022 we took part in the RoboCup Brazil Open.

Successful participation in the professional RoboCup SSL and the subsequent hardware and software improvement are not the only goals for us. Related plans also include the popularization of Robocup in Russia and the development of new solutions for educational robotics available to schools and universities. Some of the ideas described in this TDP can be used to achieve these goals.

2 Software

2.1 Transition to Python

Throughout our team's existence, all algorithms have been implemented in Matlab. This choice was justified by several reasons, which have been extensively

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described in our previous TDPs. However, starting this year, we are actively working on migrating our entire algorithmic codebase to Python. There are several reasons behind this decision, among which we would like to highlight the following:

1. Popularity: Python’s popularity significantly lowers the entry barrier for new team members, making it easier for them to contribute and get involved.
2. Object-Oriented Programming (OOP): Working with OOP in Python is much more comfortable for us compared to Functional Programming (FP) in Matlab.

2.2 Feedback Integration

One of the main issues last year was the lack of feedback integration. Although modules for its implementation were installed, we had not yet implemented through software. Integrating feedback became one of our priority tasks, and we accomplished it using the SAMR30 Module Xplained Pro. We have installed 8 modules on the control panel, each configured for different channels to process information from the robots. These modules are processed by the STM32F413VHT6 microcontroller, which then transmits the information to the Raspberry Pi.

2.3 Preface to the Software Section

This year, our goal was to significantly simplify the program structure and tactical aspects. At the Brazil Open 2022, we used the same code as in the remote RoboCup 2021, which proved to be effective. However, real matches revealed a number of serious issues that were quite challenging to fix and debug. Based on this, we made the decision to simplify many aspects, which may potentially reduce efficiency in some cases. Nevertheless, this approach will allow us to improve and upgrade the program more easily, ultimately leading us to an optimal solution.

2.4 Attack Strategy

This year, we have developed a standardized attacking formation consisting of the following elements: one robot at the center of the field controlling the ball, two robots on the sides of the opponent’s field, one robot behind the ball-carrying robot, and one robot positioned beside the ball-carrying robot (see fig. 1). We decided to abandon the non-deterministic role selection system because, despite its obvious advantages, it has several drawbacks, especially in the context of debugging.

2.5 Defense Strategy

The main idea behind our defense strategy is intercepting the opponent's passes. One of our robots approaches the opponent holding the ball with tight control. At the same time, if there is an opponent robot on our half of the field, one of our robots moves towards it to block the passing opportunity. If we have free robots available, they join the attack, creating a situation where the opponent is faced with multiple robots attempting to take the ball away (see fig. 2).

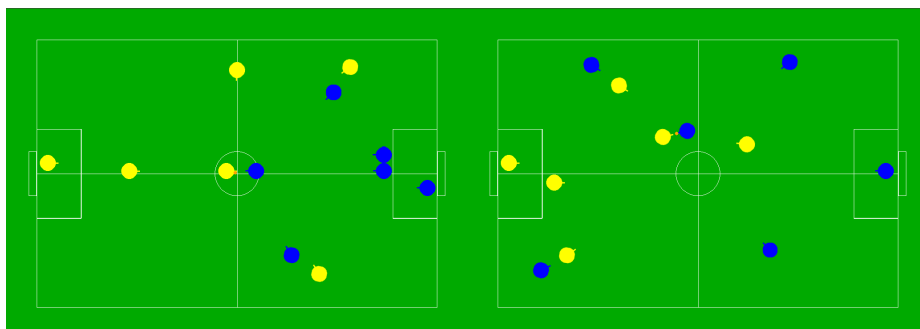


Fig. 1. Attack strategy

Fig. 2. Defence strategy

2.6 Our own simulator

We have developed a simulator that allows us to debug algorithms without the need for configuring other dependencies. This simulator is written in Python and includes its own built-in auto-referee. The main goal of the simulator is to facilitate algorithm debugging in various conditions. Even though official RoboCup simulator suits better for running full games, our solution is independent from other modules like software connecting computing server to the game server. This way our solution makes debugging tactics and algorithms easier since it can run on any computer.

2.7 Neural Networks

Inspired by the experience of other teams, we have decided to attempt training a neural network to play SSL and its individual elements within our simulator. However, we understand that the simulator is far from perfect, and we lack sufficient expertise for a full implementation. Therefore, the main objective is to explore interesting tactical solutions that the neural network can come up with and implement them in real life.

3 Hardware

The new robot (see fig. 3) design combines proven solutions from the past generation together with the introduction of innovations.

1. The torque of the drive wheels has been increased, gears and motors have been modified.
2. Kicker and dribbler were improved for better ball reception and control while moving.
3. To simplify assembly and manufacturing, as well as replacement in case of breakages, the outer elements of the protective casing have been replaced with parts designed for FDM 3D printing.



Fig. 3. Assembly of the third version of our robot

3.1 Kicker and dribbler mechanisms

The kicker assembly was modified to match the new concept of ball retention, as the dribbler and kicker affect each other. Version 2 robots had load resistors on them to discharge capacitors. The resistors generated too much heat when attached to the capacitor charging board. Also, the solenoids in the kicker assembly were connected directly to the capacitor charging board. This is not the most reliable solution, because the flexible connection is made by a single-core wire, which is not durable. We decided to make a switching board for the kicker to solve mentioned disadvantages. Here are the solutions:

1. Load resistors transferred to the switching board of the kicker are connected to the aluminium parts of the frame, which provides reliable heat dissipation.
2. The switching board will be rigidly connected to the frame, and it is supposed to be soldered with solid wires during assembly. This will increase durability as the wire will not be bent frequently during disassembly.

The switching board is also connected to the kicker capacitor charging board using a cable, which simplifies the assembly process (see fig 4).

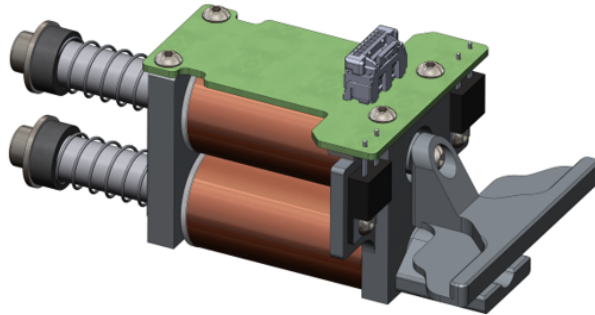


Fig. 4. Switching board for the kicker with load resistors

The dribbler is a device for holding the ball without fully securing it while the robot is moving. The new version of the dribbler has been simplified to eliminate the need for fine-tuning (see fig. 5). The new dribbler should be able to hold the ball without the need for springs. The entire dribbler is backwards compatible with the second version of the robots.

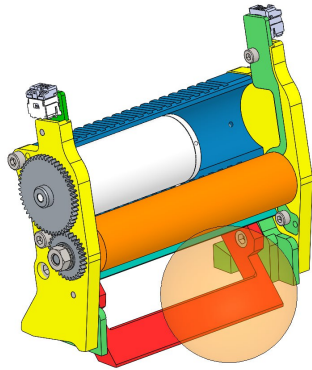


Fig. 5. Dribbler assembly

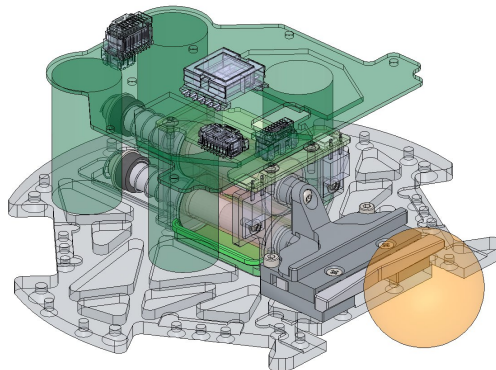


Fig. 6. Dribbler assembly

Also, to improve the quality of reception of a ball travelling at high speed, a damper has been developed that will dissipate a significant part of the kinetic energy of the ball, and thus reduce the bounce of the ball from the robot (see fig. 6).

3.2 Electronics unit

The electronics unit contains all main circuit boards and the battery. The new version integrates a camera (for more precise positioning of the robot in relation to the ball), an additional board for automatic detection of the robot's colour code and a new communication module with the base station. The battery mount has also been improved and other shortcomings of the previous model have been corrected.

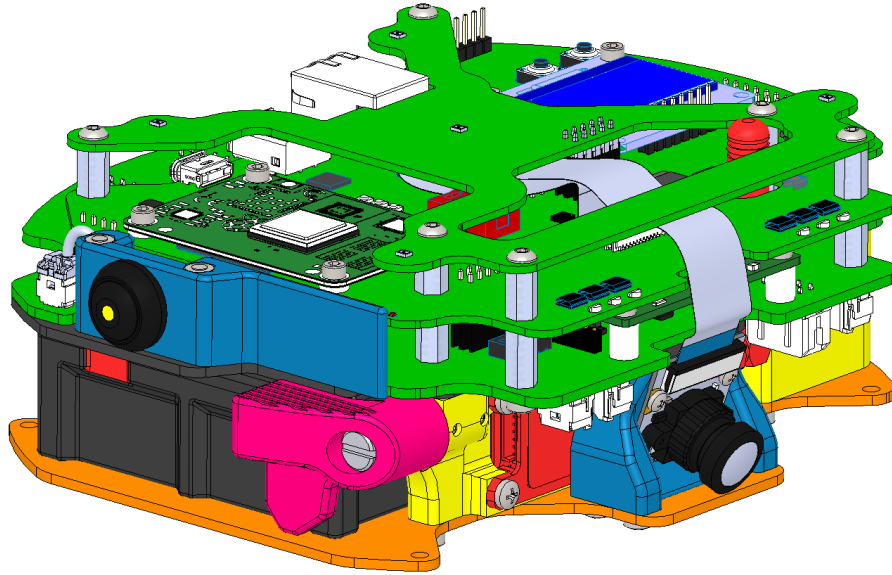


Fig. 7. Assembly of the electronics unit

The pink in the image (see fig. 7) is the new clip. It has an integrated spring to keep the battery pressed against the power connector at all times. This ensures uninterrupted battery operation during shaking and vibrations of the robot. The clamp is designed in such a way that it can be moved between positions with just one finger. Finger cutouts have also been added to the battery mount to make it easier to remove the battery.

3.3 Design modifications

A robot number and command indicator (see fig 8), located at the back under the hood, has been added to make it easier to identify the robot number while playing or debugging.

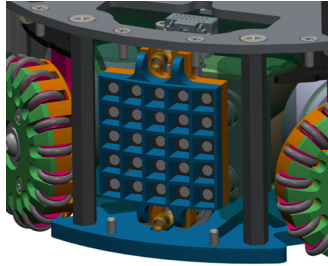


Fig. 8. Number and command indicator



Fig. 9. Member of the team blue

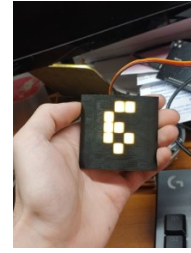


Fig. 10. Member of the team yellow

One of the main problems in robot maintenance was the protective cover, made of a bent piece of plastic, which caused inaccuracies in production. The lower guard has been redesigned to make the robot easier to service, simplify the manufacturing process and extend the life of the robot compared to the previous version of the guard.

During the operation of the robots of version 2 we appreciated the convenience of using specialized batteries in the cases with BMS. However, changing the battery and changing the communication channel with the robot required unscrewing the cover. Also, participation in LARC 2022 demonstrated the need for a solution for rapid replacement of SSL Vision tags.

A separate cover was provided for quick battery replacement, and the replacement can now be accomplished in seconds.

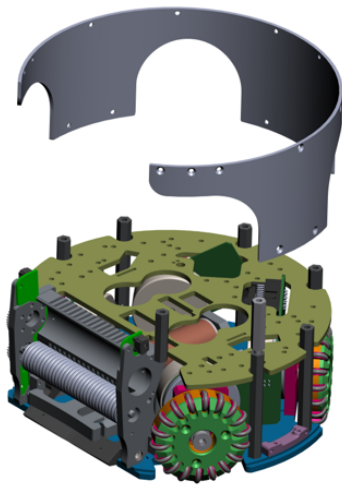


Fig. 11. Previous version of the guard

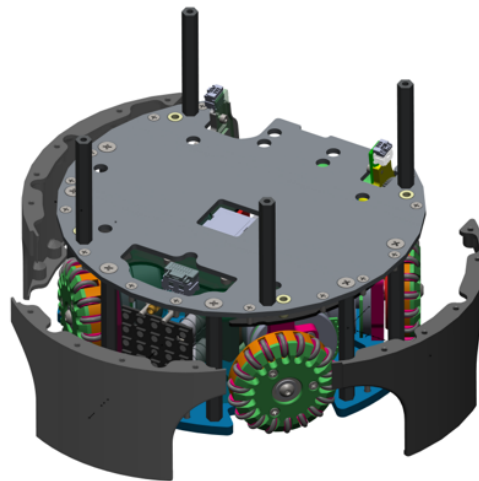


Fig. 12. Redesigned guard

The problem of changing color marks is solved with a quick release magnetized cover. 2 layers of flat magnets are able to hold sheets of the used colored paper (see fig. 13). The robot's belonging to a team and its number is detected by the colors on the back side using TCS34725 color sensors. This also speeds up the process of changing tags, as it is no longer necessary to cut and glue an even circle.

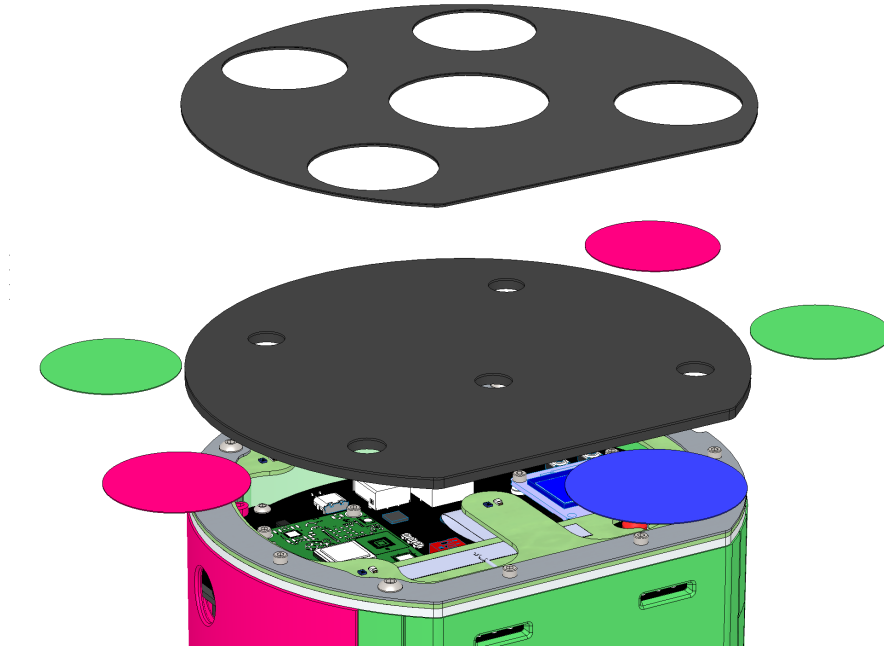


Fig. 13. Fast tags changing system

4 Electronics

The electronics of the robot underwent significant changes, they were aimed at increasing the capabilities of the robot in the game, improving the appearance and usability even for the user who is not an electronics expert.

4.1 Main controller

The main problem during the game for us was that the robot cannot adjust its actions on the field depending on the situation, and SSL Vision cannot always give a reliable picture (for example: whether the ball is exactly in the center of the dribbler to kick it as predictably as possible).

Therefore, in the new revision, a MIPI CSI camera is installed on each robot to increase the accuracy of ball and goal positioning. A Raspberry Pi Compute Module 4 added to the board to process the video stream. This single-board computer has the necessary computing power for the task, has a relatively low cost and is very compact. The only difficulty is the need for a very accurate footprint, as its connectors are very sensitive to errors in the location of the mating parts.

The solution with camera placement inside the robots is used by many strong teams, so it was decided to implement this in our robots too.

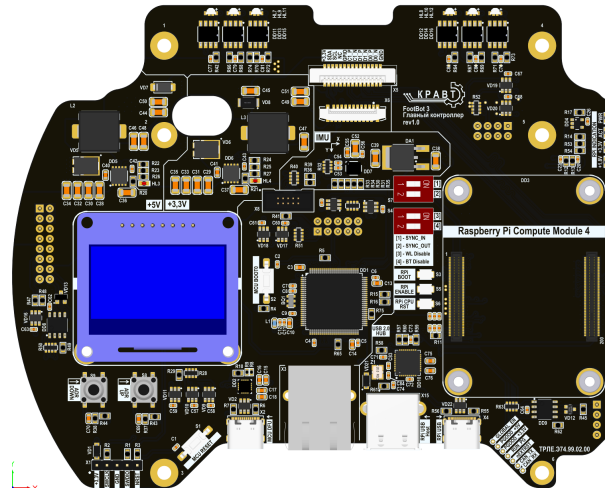


Fig. 14. Main controller

4.2 Color sensor

Compared to the last version of the robots, a board has been added to read the color markers when changing the "hat" of the robot. This will improve the user

experience in everyday life, and at competitions will reduce the chance of error, as the robot will set the correct address for signaling itself, and will not need to reset the robot as it did in the last revision.

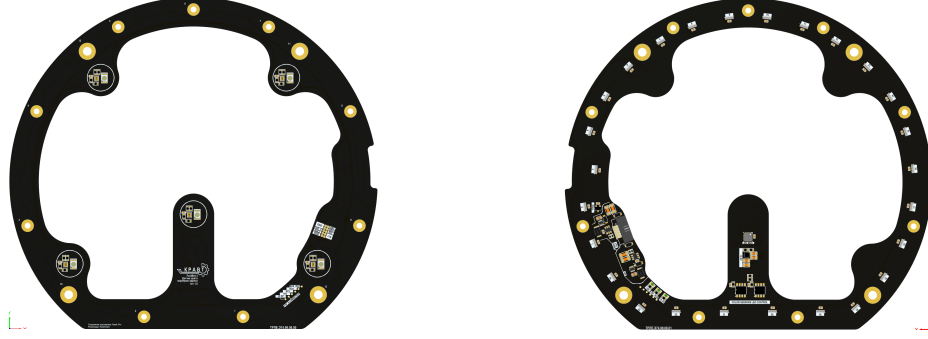


Fig. 15. Robot tag detection board (top and bottom layer)

In addition to the address recognition function, there are 24 address LEDs on the perimeter of the board, this will allow the current status of the robot to be displayed when debugging and will also improve the overall appearance of the robot.

4.3 Kicker control board

One of the main weaknesses of our team is the kicker, because compared to the robots of our colleagues, its kicking ability is significantly inferior. Therefore, we increased the kicking power of the robot by raising the voltage on the kicker board to 300 V compared to 150 V on the last version. We used the LT3751EFE boost voltage converter to charge the 390 μ F capacitors

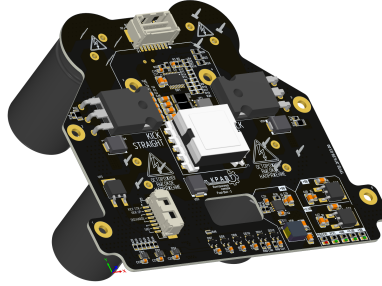


Fig. 16. Kicker boost converter

We tested this voltage on a couple of jobs at the last competition in Brazil and the result was incomparable, the robot was able to kick the ball across more of the field. Therefore, the kicker control board was modified to 300 V and the solenoid winding process was improved.

4.4 Other robot circuit boards

There are a few other boards present in the robot that have also been redesigned for the new design and corrected flaws of the last revision, but their changes are minor. The main advantage is that during the development process the principle of reducing the number of wiring in the robot was chosen, which makes it possible to speed up assembly and installation, as well as to increase reliability, since wires fail more often than plug-in connections.

The computer-robot communication unit was also redesigned, its dimensions were significantly reduced, as well as its cost. If the previous version required 220 V mains, the current version is able to work from a power bank via a USB type-C connector, which will simplify its operation.

We still use 7S Li-Po battery as a main supply of the robot, but the controller of the battery went through trouble shooting to prevent cells from dis balance.

5 Acknowledgements

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References

1. SSL Vision system, official repository, <https://github.com/RoboCup-SSL/ssl-vision>
2. Petr Konovalov, Dmitry Korolev, Dmitry Kapustin, Galina Reneva, Anastasiia Voloshina, Alexander Kalitin, and Alexander Fradkov, URoboRus 2020 Team Description Paper, https://ssl.robocup.org/wp-content/uploads/2020/03/2020_TDP_RoboRus.pdf
3. Prof. Dr. Rahib H. ABIYEV, Dr. Pavel MAKAROV, Ahmet CAGMAN, Ersin AY-TAC, Gokhan BURGE, Ali TURK, Nurullah AKKAYA, Tolga YIRTICI, Gorkem SAY, Berk YILMAZ, NEUIslanders Team Description Paper RoboCup 2019, https://ssl.robocup.org/wp-content/uploads/2019/03/2019_TDP_NEUIslanders.pdf
4. J. Almagro, C. Avidano, C. Lindbeck, J. Neiger, Z. Olkin, E. Peterson, K. Stachowicz, W. Stuckey, M. White, M. Woodward, G. P. Burdell, RoboJackets 2019 Team Description Paper, https://ssl.robocup.org/wp-content/uploads/2019/03/2019_TDP_RoboJackets.pdf
5. Tom´as Rodenas, Ricardo Alfaro, Pablo Reyes, Felipe Pinto, Maximiliano Aubel, Nicol´as Hern´andez, Pablo Yanez, Tania Barrera, Daniel Torres, Jorge Alvarez, Dami´an Quiroz, Rub´en Gonz´alez, Sysmic Robotics 2019 Team Description Paper, https://ssl.robocup.org/wp-content/uploads/2019/03/2019_TDP_SysmicRobotics.pdf
6. Kalman Filter description: https://en.wikipedia.org/wiki/Kalman_filter
7. Andre Ryll, Nicolai Ommer, Daniel Andres, Dirk Klostermann, Sebastian Nickel, Felix Pistorius, TIGERS Mannheim Team Description for RoboCup 2013 https://ssl.robocup.org/wp-content/uploads/2019/01/2013_TDP_TIGERS_Mannheim.pdf
8. Joydeep Biswas, Juan Pablo Mendoza, Danny Zhu, Phillip A. Etling, Steven Klee, Benjamin Choi, Michael Licitra, and Manuela Veloso, CMDragons 2013 Team Description https://ssl.robocup.org/wp-content/uploads/2019/01/2013_TDP_CMDragons.pdf
9. Kotaro Yasui, Yuji Nunome, Shinya Matsuoka, Yusuke Adachi, Kengo Atomi, Masahide Ito, Kunikazu Kobayashi, Kazuhito Murakami and Tadashi Naruse, RoboDragons 2013 Team Description https://ssl.robocup.org/wp-content/uploads/2019/01/2013_TDP_RoboDragons.pdf
10. TIGERS Mannheim github: <https://github.com/TIGERS-Mannheim>
11. ZJUNlict github: <https://github.com/ZJUNlict>
12. RoboDragons github: <https://github.com/RoboDragons>
13. Median Filter description: https://en.wikipedia.org/wiki/Median_filter